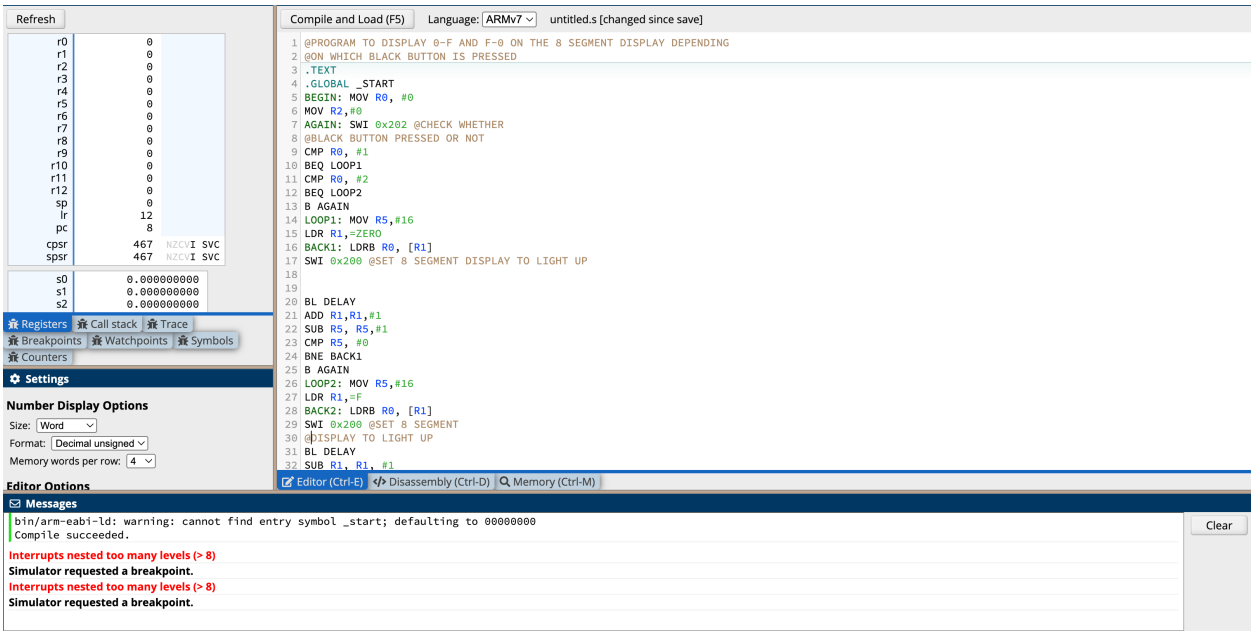
# MPCA LAB-7 SUBMISSION

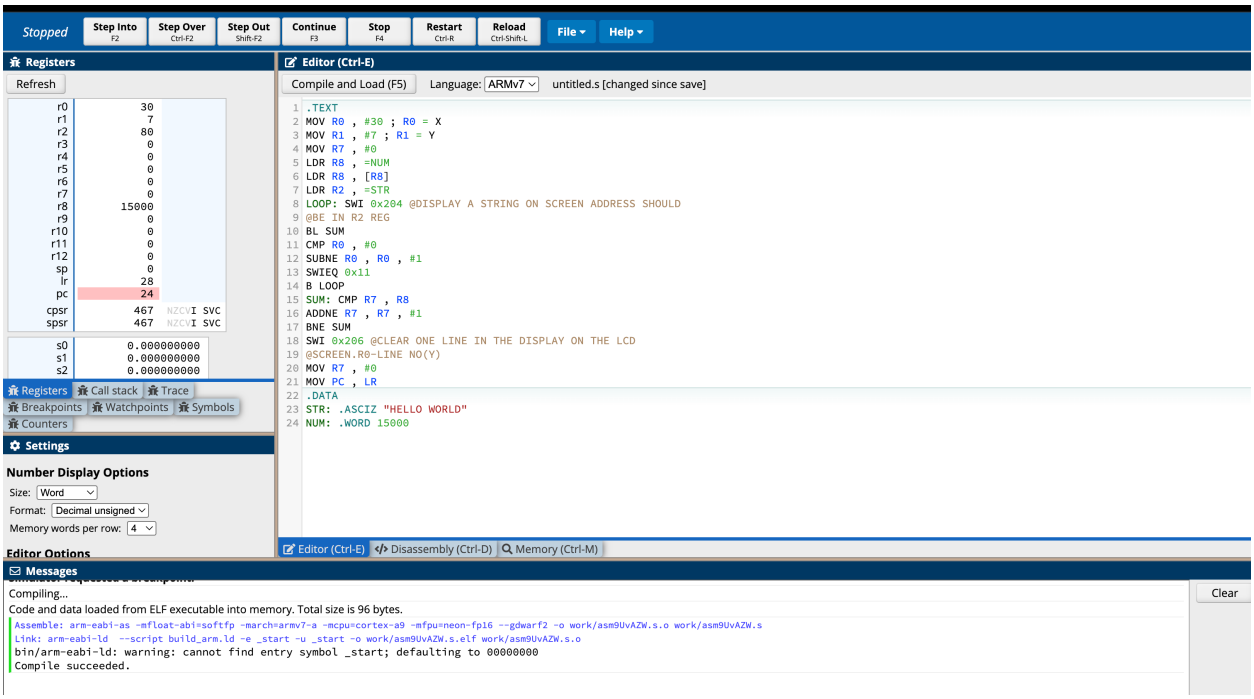
## NAME: AMARTYA SINHA

## SRN: PES1UG24CS051

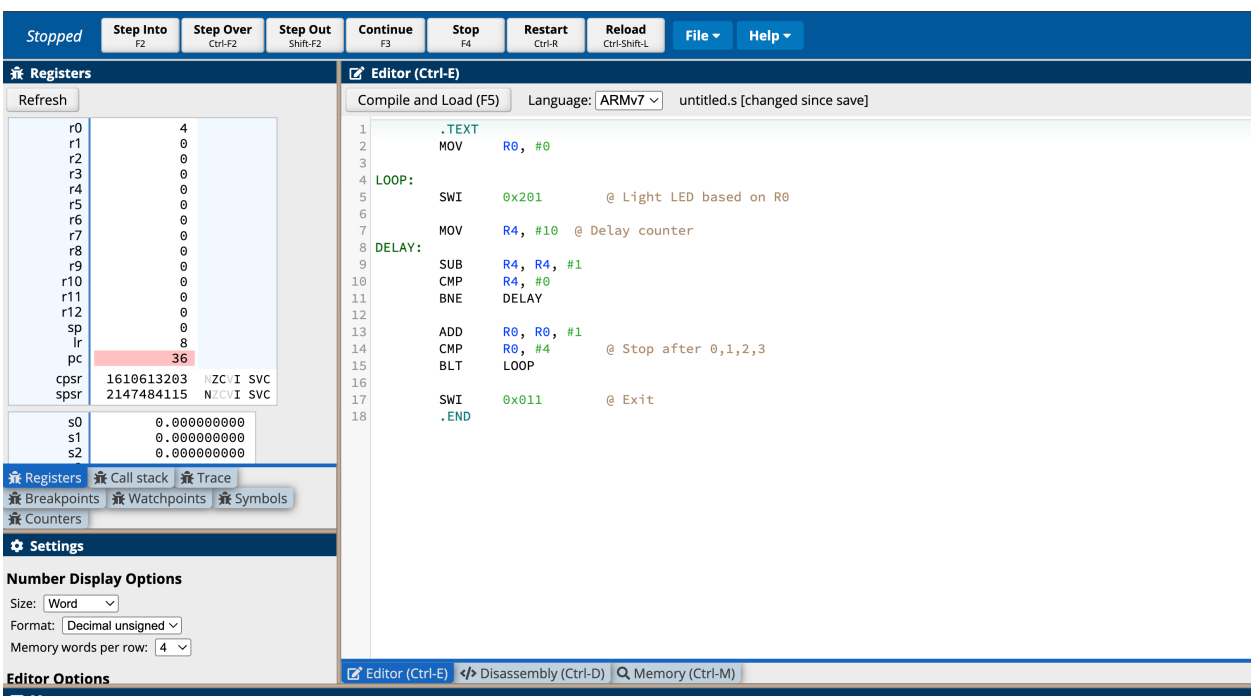
## Q1)



# Q2)


Stopped
Step Into F2
Step Over Ctrl-F2
Step Out Shift-F2
Continue F3
Stop F4
Restart Ctrl-R
Reload Ctrl-Shift-L
File
Help
Registers
Refresh
r0 30
r1 7
r2 80
r3 0
r4 0
r5 0
r6 0
r7 0
r8 15000
r9 0
r10 0
r11 0
r12 0
sp 0
lr 28
pc 24
cpsr 467 NZCVI SVC
spsr 467 NZCVI SVC
s0 0.000000000
s1 0.000000000
s2 0.000000000
Registers
Call stack
Trace
Breakpoints
Watchpoints
Symbols
Counters
Settings
Number Display Options
Size: Word
Format: Decimal unsigned
Memory words per row: 4
Editor Options
Editor (Ctrl-E)
Compile and Load (F5)
Language: ARMv7
untitled.s [changed since save]
.TEXT
MOV R0 , #30 ; R0 = X
MOV R1 , #7 ; R1 = Y
MOV R7 , #0
LDR R8 , =NUM
LDR R8 , [R8]
LDR R2 , =STR
LOOP: SWI 0x204 @DISPLAY A STRING ON SCREEN ADDRESS SHOULD
@BE IN R2 REG
BL SUM
CMP R0 , #0
SUBNE R0 , R0 , #1
SWIEQ 0x11
B LOOP
SUM: CMP R7 , R8
ADDNE R7 , R7 , #1
BNE SUM
SWI 0x206 @CLEAR ONE LINE IN THE DISPLAY ON THE LCD
@SCREEN.R0-LINE NO(Y)
MOV R7 , #0
MOV PC , LR
.DATA
STR: .ASCIZ "HELLO WORLD"
NUM: .WORD 15000
Editor (Ctrl-E)
Disassembly (Ctrl-D)
Memory (Ctrl-M)
Messages
Clear
Compiling...
Code and data loaded from ELF executable into memory. Total size is 96 bytes.
Assemble: arm-eabi-as -mfloat-abi=softfp -march=armv7-a -mcpu=cortex-a9 -mfpu=neon-fp16 --gdwarf2 -o work/asm9UvAZW.s.o work/asm9UvAZW.s
Link: arm-eabi-ld --script build_arm.ld -e _start -u _start -o work/asm9UvAZW.s.elf work/asm9UvAZW.s.o
bin/arm-eabi-ld: warning: cannot find entry symbol _start; defaulting to 00000000
Compile succeeded.

# Q3)


Stopped
Step Into F2
Step Over Ctrl-F2
Step Out Shift-F2
Continue F3
Stop F4
Restart Ctrl-R
Reload Ctrl-Shift-L
File
Help
Registers
Refresh
r0 4
r1 0
r2 0
r3 0
r4 0
r5 0
r6 0
r7 0
r8 0
r9 0
r10 0
r11 0
r12 0
sp 0
lr 8
pc 36
cpsr 1610613203 NZCVI SVC
spsr 2147484115 NZCVI SVC
s0 0.000000000
s1 0.000000000
s2 0.000000000
Registers
Call stack
Trace
Breakpoints
Watchpoints
Symbols
Counters
Settings
Number Display Options
Size: Word
Format: Decimal unsigned
Memory words per row: 4
Editor Options
Editor (Ctrl-E)
Compile and Load (F5)
Language: ARMv7
untitled.s [changed since save]
.TEXT
MOV R0, #0
LOOP:
SWI 0x201 @ Light LED based on R0
MOV R4, #10 @ Delay counter
DELAY:
SUB R4, R4, #1
CMP R4, #0
BNE DELAY
ADD R0, R0, #1
CMP R0, #4 @ Stop after 0,1,2,3
BLT LOOP
SWI 0x011 @ Exit
.END
Editor (Ctrl-E)
Disassembly (Ctrl-D)
Memory (Ctrl-M)